

EXECUTIVE SUMMARY

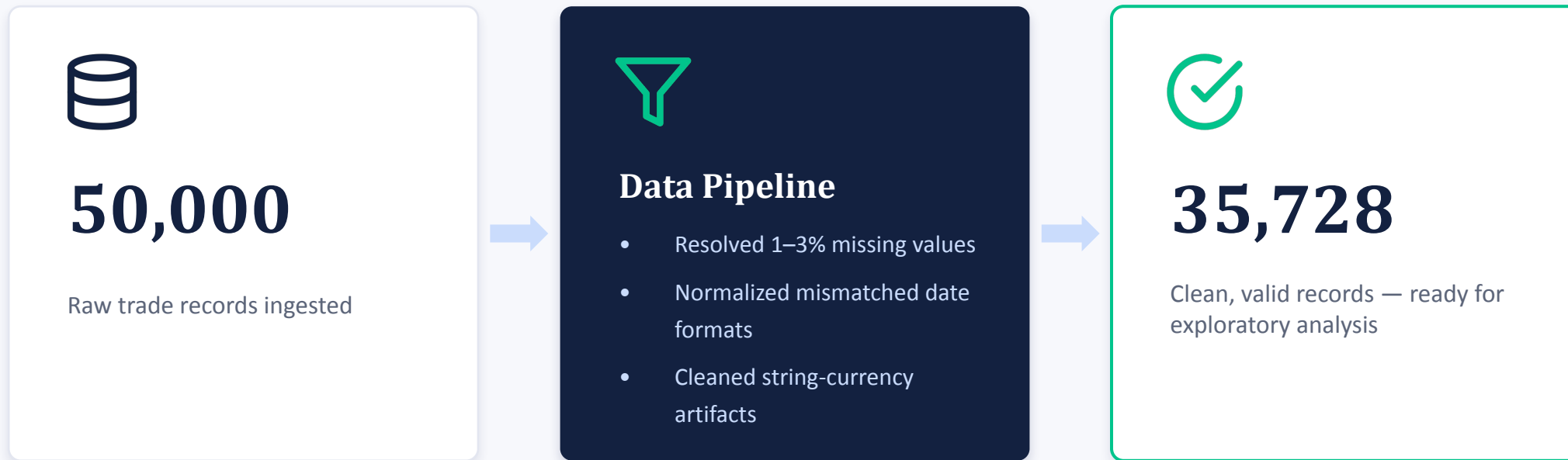
Stock Trade Data Analysis & Predictive Modeling

50,000 raw trade records • Jan 2020 – Dec 2023 • Volume-Price Trend Analysis



Project Overview & Engineering

Objective: ingest and process raw trade records to uncover market trends and evaluate volume-price relationships



71.5% of raw records retained after cleaning — a high-integrity foundation for the analysis that follows.

Key Exploratory Findings



Top Equities by Trade Volume

PG

Procter & Gamble — volume leader
737,873 shares

JPM

Closely followed

CRM

Closely followed

BAC

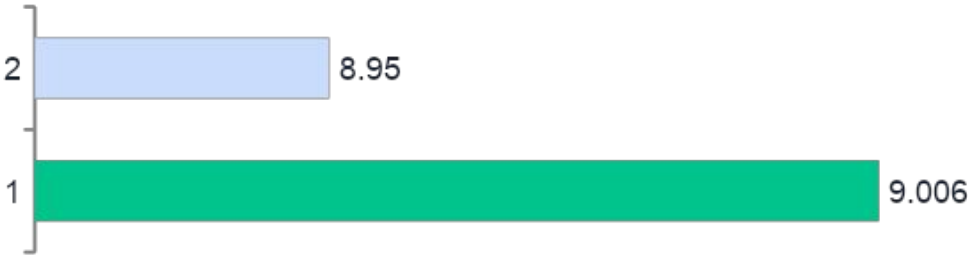
Closely followed

V

Closely followed



Brokerage Parity — Total Volume



Average Trade Size

\$501k – \$503k

Transaction averages remained tightly balanced across platforms — a sign of consistent, competitive execution industry-wide.

Predictive Modeling Insights

Ordinary Least Squares (OLS) regression of stock price on trade volume



Market Correlation

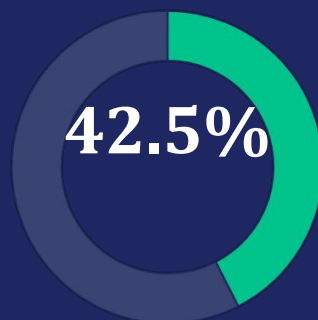
0.6518



Moderately strong positive relationship between total Trade Volume and Price.



Regression Fit (R^2)



of stock price variance is explained by trade volume in this dataset.



Statistical Significance

$p < 0.001$

HIGHLY SIGNIFICANT

Confirms trade volume is a highly significant predictor within the model parameters.